

# **An intelligent wearable insole system for machine learning-based detection of high-risk load-lifting postures**

## Table of Contents

|                                                                                                         |   |
|---------------------------------------------------------------------------------------------------------|---|
| S1. Additional Calibration Details.....                                                                 | 2 |
| S1.1 Extended FSR Sensor Calibration Data .....                                                         | 2 |
| Table S1.1. FSR Sensor Calibration Data .....                                                           | 2 |
| Figure S1.1. FSR calibration curve for all 12 sensors with error bars.....                              | 3 |
| S1.2 IMU Specifications and calibration.....                                                            | 3 |
| Table S1.2. Technical specifications of the MPU6050 sensor .....                                        | 4 |
| Figure S1.2. IMU calibration curve for X-axis .....                                                     | 4 |
| S2. Detailed Hardware Specifications .....                                                              | 4 |
| S2.1 Complete Component Specifications .....                                                            | 4 |
| Table S2.1. Component Specifications .....                                                              | 5 |
| S2.2 Circuit Design Details .....                                                                       | 5 |
| Table S2.2. ESP32 Pin Assignment .....                                                                  | 5 |
| Figure S2.1. Intelligent-insole circuit diagram.....                                                    | 6 |
| S3. Insole Construction Specifications .....                                                            | 6 |
| S3.1 Material Properties.....                                                                           | 6 |
| Table S3.1. Insole Material Properties .....                                                            | 6 |
| S3.2 Sensor Placement Precision .....                                                                   | 6 |
| Table S3.2. Anatomical Landmarks for FSR Positioning (normalized to 27cm foot).....                     | 7 |
| Figure S3.1. Anatomical foot diagram with precise FSR locations marked.....                             | 7 |
| S4. Extended Machine Learning Analysis.....                                                             | 7 |
| S4.1 Model performance.....                                                                             | 7 |
| Table S4.1. Model Performance vs. Number of Features.....                                               | 7 |
| S4.2 Cross-Validation Detailed Results .....                                                            | 8 |
| Table S4.2. 5 Fold Cross-Validation Performance with 13 features for Logistic Regression algorithm..... | 8 |
| S5. Validation Study Extended Results .....                                                             | 8 |
| S5.1 Participant-wise Performance .....                                                                 | 8 |
| Table S5.1. Participant-wise Performance .....                                                          | 8 |
| S5.2 Posture-specific Classification Performance.....                                                   | 9 |
| Table S5.2. classification accuracy by posture and load combination .....                               | 9 |

## S1. Additional Calibration Details

### S1.1 Extended FSR Sensor Calibration Data

#### Detailed Calibration Results Across Force Ranges:

In this study, a customized device (ErgoKit, Tehran, Iran) was used, developed by ErgoKit, a company experienced in manufacturing foot pressure measurement systems. The device contained 12 FSR sensors (RFP602-5KG, China) embedded to measure forces applied to different regions of the foot. For calibration, controlled forces were applied to each sensor using standard masses, which produced forces ranging from 5 N to 100 N. Each sensor underwent measurements across 20 distinct force intervals, with three repetitions per interval. Raw sensor data were collected at 5 Hz and processed in MATLAB (R2023b). The FSR's analog output was converted to Newtons following the manufacturer's specifications, using voltage mapping, resistance calculation, conductance determination, and piecewise linear force conversion based on the FSR datasheet. Normalized absolute forces were averaged over three-second intervals to obtain calibrated readings, ensuring consistent sensor measurements. A strong Pearson correlation coefficient ( $r = 0.96$ ,  $p < 0.001$ ) was observed between the calibrated FSR values and reference force gauge measurements (Table S1.1).

Table S1.1. FSR Sensor Calibration Data

| Force Range (N)  | Slope       | Intercept    | R <sup>2</sup> | RMSE (N)    | MAE (N)     |
|------------------|-------------|--------------|----------------|-------------|-------------|
| 5–10             | 0.95        | 0            | 0.88           | 1.2         | 1.0         |
| 10–15            | 0.97        | -0.85        | 0.90           | 1.1         | 0.9         |
| 15–20            | 1.00        | -0.90        | 0.91           | 1.0         | 0.8         |
| 20–25            | 1.02        | -0.95        | 0.92           | 1.0         | 0.8         |
| 25–30            | 1.04        | -1.00        | 0.93           | 0.9         | 0.7         |
| 30–35            | 1.05        | -1.05        | 0.94           | 0.9         | 0.7         |
| 35–40            | 1.06        | -1.10        | 0.94           | 0.9         | 0.7         |
| 40–45            | 1.07        | -1.15        | 0.95           | 0.8         | 0.6         |
| 45–50            | 1.08        | -1.20        | 0.95           | 0.8         | 0.6         |
| 50–55            | 1.09        | -1.25        | 0.95           | 0.8         | 0.6         |
| 55–60            | 1.10        | -1.30        | 0.96           | 0.7         | 0.5         |
| 60–65            | 1.10        | -1.35        | 0.96           | 0.7         | 0.5         |
| 65–70            | 1.11        | -1.40        | 0.96           | 0.7         | 0.5         |
| 70–75            | 1.11        | -1.45        | 0.96           | 0.7         | 0.5         |
| 75–80            | 1.12        | -1.50        | 0.96           | 0.7         | 0.5         |
| 80–85            | 1.12        | -1.55        | 0.96           | 0.7         | 0.5         |
| 85–90            | 1.13        | -1.60        | 0.96           | 0.7         | 0.5         |
| 90–95            | 1.13        | -1.65        | 0.96           | 0.7         | 0.5         |
| 95–100           | 1.14        | -1.70        | 0.96           | 0.7         | 0.5         |
| <b>Overall</b>   | <b>1.03</b> | <b>-0.94</b> | <b>0.92</b>    | <b>2.79</b> | <b>2.21</b> |
| <b>Pearson r</b> |             |              | <b>0.96</b>    |             |             |

#### Non-linearity Characterization:

- Below 10N: Average overestimation of  $8.3\% \pm 2.1\%$
- 10-50N: Linear response within  $\pm 3\%$  of reference

- Above 50N: Slight underestimation of  $4.7\% \pm 1.8\%$
- Hysteresis: Maximum 2.3% of full scale

The force–response behavior of the FSR sensors in Figure S1.1 showed slight non-linearity across different force ranges. At low forces (5–10 N), the sensors slightly underestimated the applied force with a slope of 0.95 and  $R^2 = 0.88$ , while in mid-ranges (10–50 N) the response became more linear (slopes approaching 1.00–1.08,  $R^2 = 0.90$ –0.95). At higher forces (>50 N), the sensors slightly overestimated the applied force, with slopes increasing up to 1.14 and  $R^2$  remaining stable at 0.96. The intercept values gradually decreased from 0 to  $-1.70$  N as the force range increased. Overall, the sensor array exhibited good linearity with an average slope of 1.03, overall  $R^2 = 0.92$ , RMSE = 2.79 N, MAE = 2.21 N, and strong correlation between applied and measured forces (Pearson  $r = 0.96$ ).

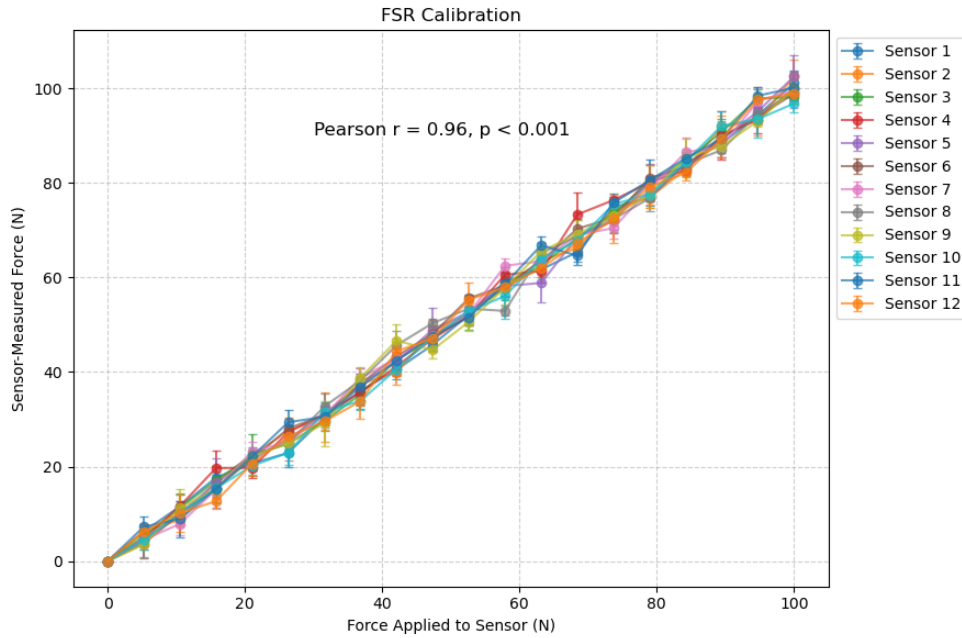


Figure S1.1. FSR calibration curve for all 12 sensors with error bars

## S1.2 IMU Specifications and calibration

### Sensor Specifications

Trunk flexion angle measurements were performed using the GY-52 , MPU605 IMU sensor (InvenSense), which integrates a 3-axis gyroscope and a 3-axis accelerometer. With Gyroscope drift:  $-0.013^\circ/\text{s}$  per  $^\circ\text{C}$ , Accelerometer offset:  $0.0008$  g per  $^\circ\text{C}$  and Operating range validated in  $15$ – $35^\circ\text{C}$ , other key specifications are summarized in Table S1.2.

Table S1.2. Technical specifications of the MPU6050 sensor

| Parameter                       | Specification                              |
|---------------------------------|--------------------------------------------|
| Communication Interface         | I2C (up to 400 kHz)                        |
| Supply Voltage                  | 2.375 – 3.46 V                             |
| Accelerometer Range             | $\pm 2g, \pm 4g, \pm 8g, \pm 16g$          |
| Gyroscope Range                 | $\pm 250, \pm 500, \pm 1000, \pm 2000$ °/s |
| Operating Temperature           | -40°C to +85°C                             |
| Current Consumption             | ~3.9 mA (active mode)                      |
| FIFO Buffer Size                | 1024 bytes                                 |
| Digital Motion Processing (DMP) | Supported                                  |

## Calibration and Validation

The MPU6050 was calibrated against a clinical goniometer over the range of 0–180° for the pitch axis. Calibration demonstrated a strong linear correlation between IMU-derived and goniometer-measured angles ( $r=0.99$ ,  $p<0.001$ ). Bland-Altman analysis indicated a mean difference of  $0.2 \pm 0.8$  degrees and 95% limits of agreement from  $-1.37^\circ$  to  $+1.77^\circ$ , confirming the sensor's suitability for precise trunk angle measurement (Figure S1.2).

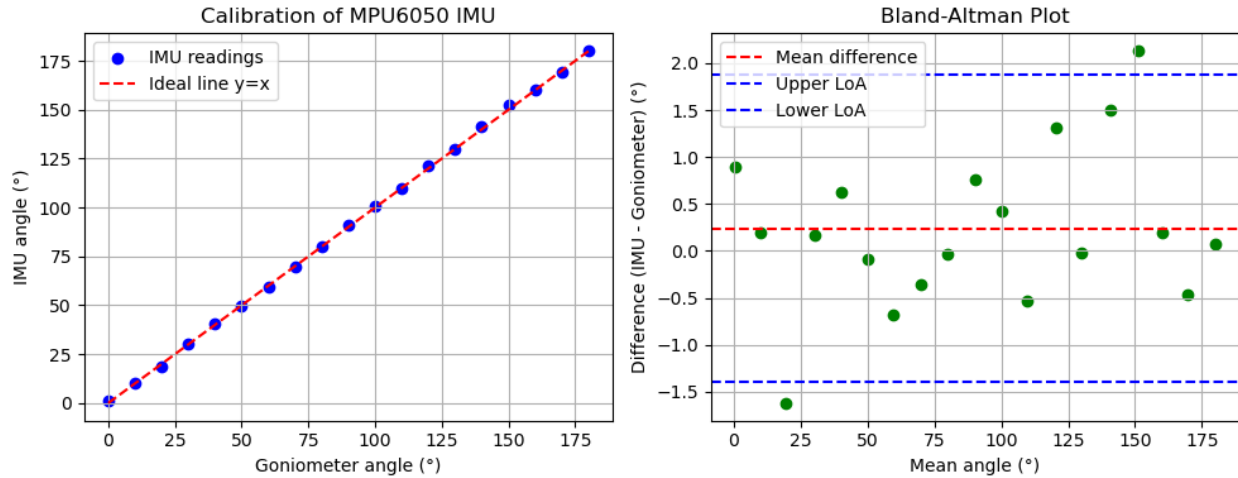


Figure S1.2. IMU calibration curve for X-axis

## S2. Detailed Hardware Specifications

### S2.1 Complete Component Specifications

The intelligent insole system was designed with a compact structure optimized for real-time sensing and feedback. It includes an ESP32-WROOM-32 microcontroller for data processing and communication, an MPU6050 IMU for measuring trunk motion, and multiple FSR sensors embedded in the insole to record plantar pressure. A CD74HC4067 multiplexer manages multiple sensor inputs, supported by an 18650 Lithium battery and a voltage regulator for stable power.

supply. Real-time feedback to the user is provided through an integrated LED and buzzer module. The detailed specifications of all components are presented in Table S2.1.

Table S2.1. Component Specifications

| Component         | Model          | Specifications                            | Power Consumption    |
|-------------------|----------------|-------------------------------------------|----------------------|
| Microcontroller   | ESP32-WROOM-32 | 240 MHz dual-core, 520 KB SRAM            | 80 mA at 3.3V        |
| IMU               | MPU6050        | $\pm 2g$ , $\pm 250^\circ/s$ , 16-bit ADC | 3.8 mA at 3.3V       |
| FSR               | 602-5K         | 0.2-20N range, 5mm diameter               | Passive              |
| Multiplexer       | CD74HC4067     | 16-channel, $2\Omega$ ON resistance       | 0.16 $\mu A$ at 3.3V |
| Battery           | 18650 Lithium  | 2600mAh, 3.7V nominal                     | -                    |
| Voltage Regulator | AMS1117-3.3    | 1A output, 1.3V dropout                   | 5 mA quiescent       |
| Buzzer            | KY-006         | 85dB at 10cm, 2-5 kHz                     | 30 mA at 3.3V        |
| LED (Green)       | 5mm diffused   | 20 mcd, 525 nm                            | 20 mA at 2.1V        |
| LED (Red)         | 5mm diffused   | 20 mcd, 625 nm                            | 20 mA at 2.0V        |

## S2.2 Circuit Design Details

The circuit was organized in a compact layout, ensuring stable data transmission between sensors and the microcontroller while maintaining efficient power distribution and reliable system performance (Table S2.2 and Figure S2.1).

Table S2.2. ESP32 Pin Assignment

| Function                   | ESP32 Pin | Additional Details    |
|----------------------------|-----------|-----------------------|
| <b>I2C Communication</b>   |           |                       |
| MPU6050 SDA                | GPIO 21   | 4.7k $\Omega$ pull-up |
| MPU6050 SCL                | GPIO 22   | 4.7k $\Omega$ pull-up |
| <b>Multiplexer Control</b> |           |                       |
| MUX S0                     | GPIO 13   | LSB select            |
| MUX S1                     | GPIO 12   | -                     |
| MUX S2                     | GPIO 14   | -                     |
| MUX S3                     | GPIO 27   | MSB select            |
| MUX SIG                    | GPIO 34   | ADC1 CH6              |
| MUX EN                     | GPIO 32   | Active LOW            |
| <b>Feedback Outputs</b>    |           |                       |
| Green LED                  | GPIO 25   | 330 $\Omega$ limiting |
| Red LED                    | GPIO 26   | 330 $\Omega$ limiting |
| Buzzer                     | GPIO 33   | PWM capable           |

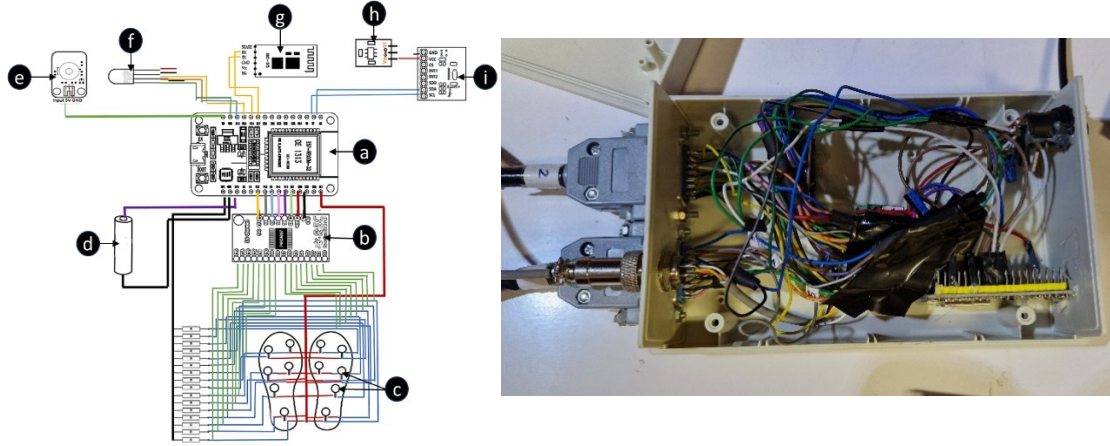


Figure S2.1. Intelligent insole circuit diagram. (a) ESP32 microcontroller; (b) analog multiplexer; (c) FSR sensors; (d) 18650 Lithium battery; (e) buzzer; (f) LED indicator; (g) Bluetooth module; (h) voltage regulator; (i) IMU.

### S3. Insole Construction Specifications

#### S3.1 Material Properties

The intelligent insole was constructed as a multilayer system designed to balance durability, comfort, and sensor stability. The base layer made of wooden fiberboard provides structural support and a stable surface for sensor placement. A cushioning layer of EVA foam ensures comfort and shock absorption during movement. The top cover, made of a flexible TPU sheet, enhances durability and user comfort while protecting embedded components. All layers were securely bonded using a thin polyurethane adhesive, maintaining flexibility and uniform pressure distribution across the insole. The material properties of each layer are summarized in Table S3.1.

Table S3.1. Insole Material Properties

| Layer      | Material          | Thickness | Density                | Shore Hardness |
|------------|-------------------|-----------|------------------------|----------------|
| Base       | Wooden Fiberboard | 3 mm      | 650 kg/m <sup>3</sup>  | -              |
| Cushioning | EVA Foam          | 5 mm      | 95 kg/m <sup>3</sup>   | 45A            |
| Top Cover  | TPU Sheet         | 1 mm      | 1200 kg/m <sup>3</sup> | 85A            |
| Adhesive   | Polyurethane      | 0.2 mm    | -                      | -              |

#### S3.2 Sensor Placement Precision

The insole was designed to ensure precise and anatomically consistent placement of the twelve FSR sensors across the plantar surface. Each sensor was aligned according to identifiable anatomical landmarks normalized to a 27 cm foot length, ensuring consistent mapping across participants. Sensors were positioned to capture pressure distribution from key load-bearing regions, including the heel, midfoot, metatarsal heads, and toes. This layout enabled accurate detection of balance shifts and load transfer during lifting tasks. The detailed sensor locations are listed in Table S3.2, and their spatial arrangement is illustrated in Figure S3.1.

Table S3.2. Anatomical Landmarks for FSR Positioning (normalized to 27cm foot)

| Sensor  | Anatomical Location  | Position (% of foot length) | Lateral Position |
|---------|----------------------|-----------------------------|------------------|
| FSR1/7  | Heel centrum         | 50% from heel               | Midline          |
| FSR2/8  | Navicular tuberosity | 65% from heel               | 30% lateral      |
| FSR3/9  | 1st metatarsal head  | 78% from heel               | 25% medial       |
| FSR4/10 | 5th metatarsal head  | 78% from heel               | 25% lateral      |
| FSR5/11 | Hallux IP joint      | 92% from heel               | 20% medial       |
| FSR6/12 | 3rd toe MTP joint    | 92% from heel               | Midline          |

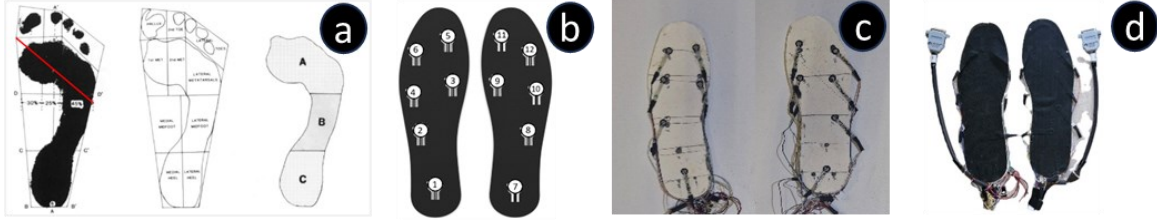


Figure S3.1. Anatomical foot diagram with precise FSR locations marked

## S4. Extended Machine Learning Analysis

### S4.1 Model performance

Table S4.1 presents the classification performance of different machine learning models as the number of features increased through incremental feature selection. The models were evaluated using AUC, specificity, sensitivity, and overall accuracy. Both Support Vector Machine (SVM) and Logistic Regression (LR) showed steady performance improvement with additional features, indicating the contribution of plantar pressure sensors to classification accuracy. The Logistic Regression model, incorporating trunk angle and twelve FSR inputs, achieved the highest performance with an AUC of 0.94 and an accuracy of 91.17%, demonstrating its robustness and suitability for real-time posture classification in the intelligent insole system.

Table S4.1. Model Performance vs. Number of Features

| Features |            | Model                  | Accuracy | Sensitivity | Specificity | AUC  |
|----------|------------|------------------------|----------|-------------|-------------|------|
| 1        | TrunkAngle | Logistic Regression    | 0        | 0           | 0           | 0    |
| 2        | +FSR 7     | Support Vector Machine | 72.94%   | 65.62%      | 82.43%      | 0.75 |
| 3        | +FSR 1     | Support Vector Machine | 79.41%   | 71.87%      | 89.18%      | 0.77 |
| 4        | +FSR 5     | Support Vector Machine | 72.94%   | 71.87%      | 74.32%      | 0.81 |
| 5        | +FSR 6     | Logistic Regression    | 72.94%   | 71.87%      | 74.32%      | 0.79 |
| 6        | +FSR 11    | Support Vector Machine | 80.58%   | 86.45%      | 72.97%      | 0.82 |
| 7        | +FSR 12    | Logistic Regression    | 80.58%   | 81.25%      | 79.72%      | 0.87 |
| 8        | +FSR 4     | Support Vector Machine | 82.35%   | 83.33%      | 81.08%      | 0.87 |
| 9        | +FSR 10    | Support Vector Machine | 85.29%   | 88.54%      | 81.08%      | 0.88 |
| 10       | +FSR 3     | Logistic Regression    | 84.70%   | 82.29%      | 87.83%      | 0.90 |
| 11       | +FSR 9     | Support Vector Machine | 81.76%   | 82.29%      | 81.08%      | 0.90 |
| 12       | +FSR 2     | Support Vector Machine | 85.29%   | 88.54%      | 81.08%      | 0.91 |
| 13       | +FSR 8     | Logistic Regression    | 91.17%   | 93.75%      | 87.83%      | 0.94 |

## S4.2 Cross-Validation Detailed Results

Table S4.2 presents the detailed performance metrics of the Logistic Regression model using 13 selected features across 5-fold cross-validation. For each fold, we report class-specific F1-score, recall, and precision for both high-risk (Label 2) and low-risk (Label 1) postures, as well as the overall training and validation accuracy. The model demonstrates consistently high performance, with mean F1-scores of 0.90 and 0.92 for high-risk and low-risk postures, respectively, and mean validation and training accuracies of 91.1% and 92.3%. The standard deviations are minimal ( $\leq 0.009$  for F1, recall, and precision;  $\leq 0.5\%$  for validation accuracy), indicating low variability across folds and stable generalization of the model.

Table S4.2. 5 Fold Cross-Validation Performance with 13 features for Logistic Regression algorithm

| Fold        | Train Accuracy | Validation Accuracy | Low-risk posture (Label 1) |              |              | High-risk posture (Label 2) |              |              |
|-------------|----------------|---------------------|----------------------------|--------------|--------------|-----------------------------|--------------|--------------|
|             |                |                     | Precision                  | Recall       | F1-score     | Precision                   | Recall       | F1-score     |
| 1           | 92.1%          | 90.8%               | 0.90                       | 0.93         | 0.91         | 0.88                        | 0.91         | 0.89         |
| 2           | 91.9%          | 91.3%               | 0.91                       | 0.95         | 0.93         | 0.89                        | 0.92         | 0.90         |
| 3           | 92.5%          | 90.5%               | 0.89                       | 0.94         | 0.91         | 0.90                        | 0.91         | 0.90         |
| 4           | 92.3%          | 91.7%               | 0.90                       | 0.93         | 0.91         | 0.88                        | 0.93         | 0.90         |
| 5           | 92.7%          | 91.2%               | 0.90                       | 0.95         | 0.92         | 0.89                        | 0.93         | 0.91         |
| <b>Mean</b> | <b>92.3%</b>   | <b>91.1%</b>        | <b>0.90</b>                | <b>0.94</b>  | <b>0.92</b>  | <b>0.89</b>                 | <b>0.92</b>  | <b>0.90</b>  |
| <b>SD</b>   | <b>0.3%</b>    | <b>0.5%</b>         | <b>0.007</b>               | <b>0.009</b> | <b>0.008</b> | <b>0.008</b>                | <b>0.008</b> | <b>0.007</b> |

## S5. Validation Study Extended Results

### S5.1 Participant-wise Performance

Table S5.1 summarizes the classification performance of the model for each participant. For every participant, true positives (low-risk postures correctly identified), true negatives (high-risk postures correctly identified), false positives (high-risk  $\rightarrow$  low-risk), and false negatives (low-risk  $\rightarrow$  high-risk) are reported alongside overall accuracy. Across the 10 participants, the model consistently demonstrates high accuracy, with individual values ranging from 84% to 88% and an overall accuracy of 88.4%. These results indicate reliable participant-wise performance and confirm that the model generalizes well across different individuals.

Table S5.1. Participant-wise Performance

| Participant | True Positives (Low-risk) | True Negatives (High-risk) | False Positives (High-risk $\rightarrow$ Low-risk) | False Negatives (Low-risk $\rightarrow$ High-risk) | Accuracy |
|-------------|---------------------------|----------------------------|----------------------------------------------------|----------------------------------------------------|----------|
| P1          | 7                         | 15                         | 2                                                  | 1                                                  | 88%      |
| P2          | 6                         | 16                         | 2                                                  | 1                                                  | 88%      |
| P3          | 7                         | 15                         | 1                                                  | 2                                                  | 88%      |



|              |           |            |           |           |              |
|--------------|-----------|------------|-----------|-----------|--------------|
| P4           | 6         | 16         | 2         | 1         | 88%          |
| P5           | 6         | 15         | 2         | 2         | 84%          |
| P6           | 7         | 15         | 1         | 2         | 88%          |
| P7           | 6         | 16         | 2         | 1         | 88%          |
| P8           | 6         | 16         | 2         | 1         | 88%          |
| P9           | 7         | 15         | 1         | 2         | 88%          |
| P10          | 6         | 16         | 2         | 1         | 88%          |
| <b>Total</b> | <b>64</b> | <b>155</b> | <b>17</b> | <b>14</b> | <b>88.4%</b> |

## S5.2 Posture-specific Classification Performance

Table S5.2 reports the classification accuracy of the model for each posture across varying load conditions. The model achieved 100% accuracy for several postures, including A1 under both 6 kg and 12 kg loads, as well as the neutral posture, while other postures exhibited accuracies ranging from 70% to 100%. These results demonstrate that the model effectively discriminates between postures under different loading scenarios, with minor reductions in accuracy reflecting the increased complexity of certain posture-load combinations.

Table S5.2. classification accuracy by posture and load combination

| Posture | Load (kg) | Correct / Total | Accuracy |
|---------|-----------|-----------------|----------|
| A1      | 6         | 10/10           | 100%     |
| A1      | 12        | 10/10           | 100%     |
| A2      | 6         | 9/10            | 90%      |
| A2      | 12        | 10/10           | 100%     |
| A3      | 6         | 9/10            | 90%      |
| A3      | 12        | 10/10           | 100%     |
| A4      | 6         | 9/10            | 90%      |
| A4      | 12        | 9/10            | 90%      |
| B1      | 6         | 7/10            | 70%      |
| B1      | 12        | 9/10            | 90%      |
| B2      | 6         | 8/10            | 80%      |
| B2      | 12        | 9/10            | 90%      |
| B3      | 6         | 7/10            | 70%      |
| B3      | 12        | 8/10            | 80%      |
| B4      | 6         | 7/10            | 70%      |
| B4      | 12        | 8/10            | 80%      |
| C1      | 6         | 8/10            | 80%      |
| C1      | 12        | 9/10            | 90%      |
| C2      | 6         | 9/10            | 90%      |
| C2      | 12        | 10/10           | 100%     |
| C3      | 6         | 8/10            | 80%      |
| C3      | 12        | 9/10            | 90%      |
| C4      | 6         | 9/10            | 90%      |
| C4      | 12        | 9/10            | 90%      |
| Neutral | N/A       | 10/10           | 100%     |